IMPLEMENTATION & REVIEW PLAN · V2.0 · COURSE FORMAT

Build, Review & Integration Protocol

MPSTME GUARDIAN — NMIMS Main Building scene

Four members · one shared control file · one integrator on `main`

> *The course format puts all four members' code in one file. That is efficient to read and dangerous to edit in parallel. This plan makes it safe: every line has one owner, every pull request is checked against those lines, and Muaaz is the only person who merges.*

COMPANION TO	Product Requirements Document v2.0
REPOSITORY	`Group_05_Muaaz_Aditya_Zaid_Soumya/`
INTEGRATOR	E043 Muaaz Mohammed Iqbal Shaikh — owns `main`; pulls, gates, reviews and merges every branch
BRANCHES	As assigned in `docs/TEAM_ROSTER.json`
GATE	`src/check_region_ownership.py` — built and self-tested
DATE	8 October 2026

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1 Where the Project Stands

The scene, the airframes, the flight stack and every integrity gate are built and measured. What remains is each member's own region — and those can now start in parallel on day one.

ITEM	OWNER	STATUS	EVIDENCE
NMIMS Main Building scene from OSM	E043	DONE	342 m ring, 42 context buildings, real-vs-sim figure
Scaffold model fixes (camera, yaw, I, f)	E043	DONE	`test_env.py` passes
Physics flight loop + cascade + mixer	E043	DONE	±0.033 m altitude, 0 collisions
Layered separation	E043	DONE	head-on 3.29 m vs 0.40 m APF-only
Benchmark runner + provenance	E043	DONE	smoke batch, 3 trials
Read-only figure generator	E043	DONE	refuses scaffold CSV, exit 2
Region ownership gate	E043	DONE	self-test: allow own region, block cross-region
HSV detector, pose-aware projection, IoU	E051	OPEN	stub raises `NotImplementedError`
Polygon geofence, hysteresis FSM, RTL triggers, coverage	E075	OPEN	scaffold code still uses old-scene bounds
Guard comparator, economics from CSV	E077	OPEN	constants still typed in

2 Team, Branches & Ownership

ROLL	SAP ID	MEMBER	COURSE BRANCH	OWNS
E043	70362400030	Muaaz M. I. Shaikh LEAD · INTEGRATOR	`feat/e043-lead-uav-flight-dyna`	`models/` ; E043 regions + integration loop; `run_benchmark.py` ; `generate_paper_figures.py` ; `main`
E051	70362400067	Aditya Rajkumar VISION	`feat/e051-computer-vision-open`	E051 region; `analytics/vision_*` ; `tests/test_e051_*`
E075	70362400003	Zaid Rezaur Rahman GEOFENCE	`feat/e075-restricted-zone-geo-`	E075 region; `analytics/coverage_*` ; `tests/test_e075_*`

ROLL	SAP ID	MEMBER	COURSE BRANCH	OWNS
E077	70362400080	Soumya S. Ranasingh CSBS	`feat/e077-csbs-campus-security`	`campus_security_economics.py` ; `analytics/guard_patrol_baseline*` ; `tests/test_e077_*`

2.1 Regions inside the shared control file

```
# ======================================================================
# >>> E051 ADITYA -- COMPUTER VISION  >>>
# ======================================================================
    def detect_intruder_hsv(self, rgb_frame): ...
    def inverse_pinhole_projection(self, u_px, v_px, uav_pos, ...): ...
# ======================================================================
# <<< E051 ADITYA <<<
# ======================================================================
```

Lines between a member's markers are theirs. Lines outside any region — imports, the class constructor, `main()` — and the marker lines themselves belong to E043 as shared plumbing. The gate reads these markers from both the base and the branch version of the file, so a change that moves a marker is caught too.

THE RULE THAT PROTECTS EVERYONE'S VIVA MARKS

Muaaz reviews, requests changes and merges — he does not write inside anyone else's region. If a member is stuck, the fix is a pairing session or a smaller scope, never a commit under the wrong name. The viva asks each student to modify and re-run their own code live.

3 Repository Layout

The course tree is preserved. Additions are marked **+**; nothing from the scaffold was deleted.

```
Group_05_Muaaz_Aditya_Zaid_Soumya/
├── README.md                                  tree + scene section updated            [E043]
├── RESEARCH_AND_IMPLEMENTATION_GUIDE.md       FROZEN — course-provided
├── docs/
│   ├── TEAM_ROSTER.json                       LinkedIn URL goes here                  [E043]
│   ├── LITERATURE_REVIEW_AND_FOUNDATIONAL_PAPERS.md                                   [all]
│   ├── RESEARCH_PAPER_MANUSCRIPT_BLUEPRINT.md the 4-page paper                        [all]
│ + ├── planning/                              PRD v2.0 + this plan                    [E043]
│   └── figures/
│       ├── figure1..3_*.png                   regenerated from data only
│     + └── scene/                             control_room, tour_viewpoint, real-vs-sim
├── models/
│   ├── campus_perimeter_patrol.xml            airframes, cameras, intruder, zone      [E043]
│ + ├── nmims_main_campus.xml                  GENERATED — do not hand-edit
│ + └── scene_source/                          generator, OSM extract, manifest        [E043]
├── src/
│   ├── test_env.py                            unchanged
│   ├── aerial_patrol_swarm.py                 4 regions: E043 · E051 · E075 · E043-loop
│ + └── check_region_ownership.py              the PR gate                             [E043]
│ + tests/
│ +   └── test_e043_separation.py              head-on regression                      [E043]
└── analytics/
    ├── campus_security_economics.py                                                    [E077]
  + ├── run_benchmark.py                       only producer of the benchmark CSV      [E043]
    ├── generate_paper_figures.py              rewritten: read-only, provenance gate   [E043]
    ├── campus_patrol_benchmark.csv            scaffold placeholder until the first real batch
  + └── runs/                                  per-run telemetry (gitignored)
```

4 Git Workflow

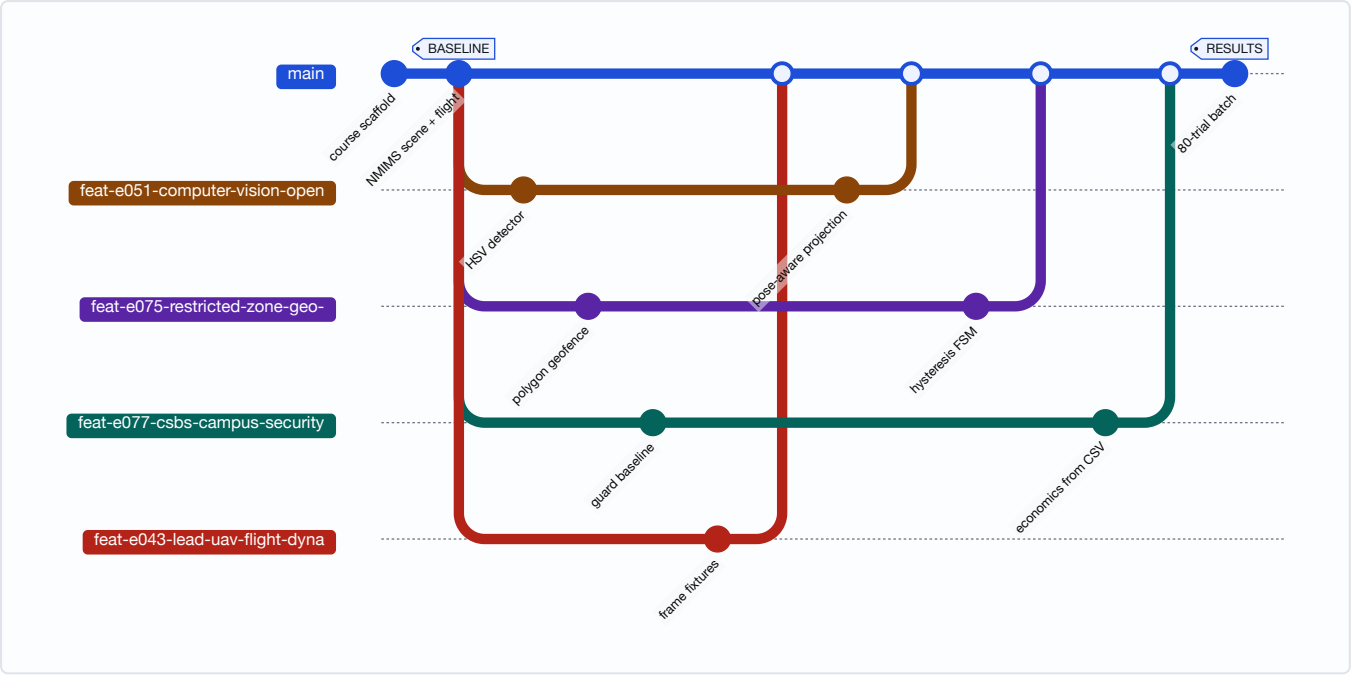


Figure 1 — Branch model with the course-assigned branch names. (Mermaid cannot draw "/" in a branch label, so it is shown as "-".) Every merge is a reviewed pull request that has passed the gates in §13. Branches are rebased on `main` daily so they never drift for long.

RULE	WHY
Each member sets their own `user.name` / `user.email`	The viva audits individual commit history.
Merge commits, never squash	Squashing reattributes a member's commits to whoever pressed the button.
Commit subject `<area>: <imperative>`	`detector: reject blobs under 40 px` reads cleanly in `git log`.
Commit body cites the FR	"Satisfies FR-16" links code to requirement for the reviewer.
Nobody pushes to `main` except Muaaz	One integrator, one place where things can break.

5 Muaaz's Integration Procedure

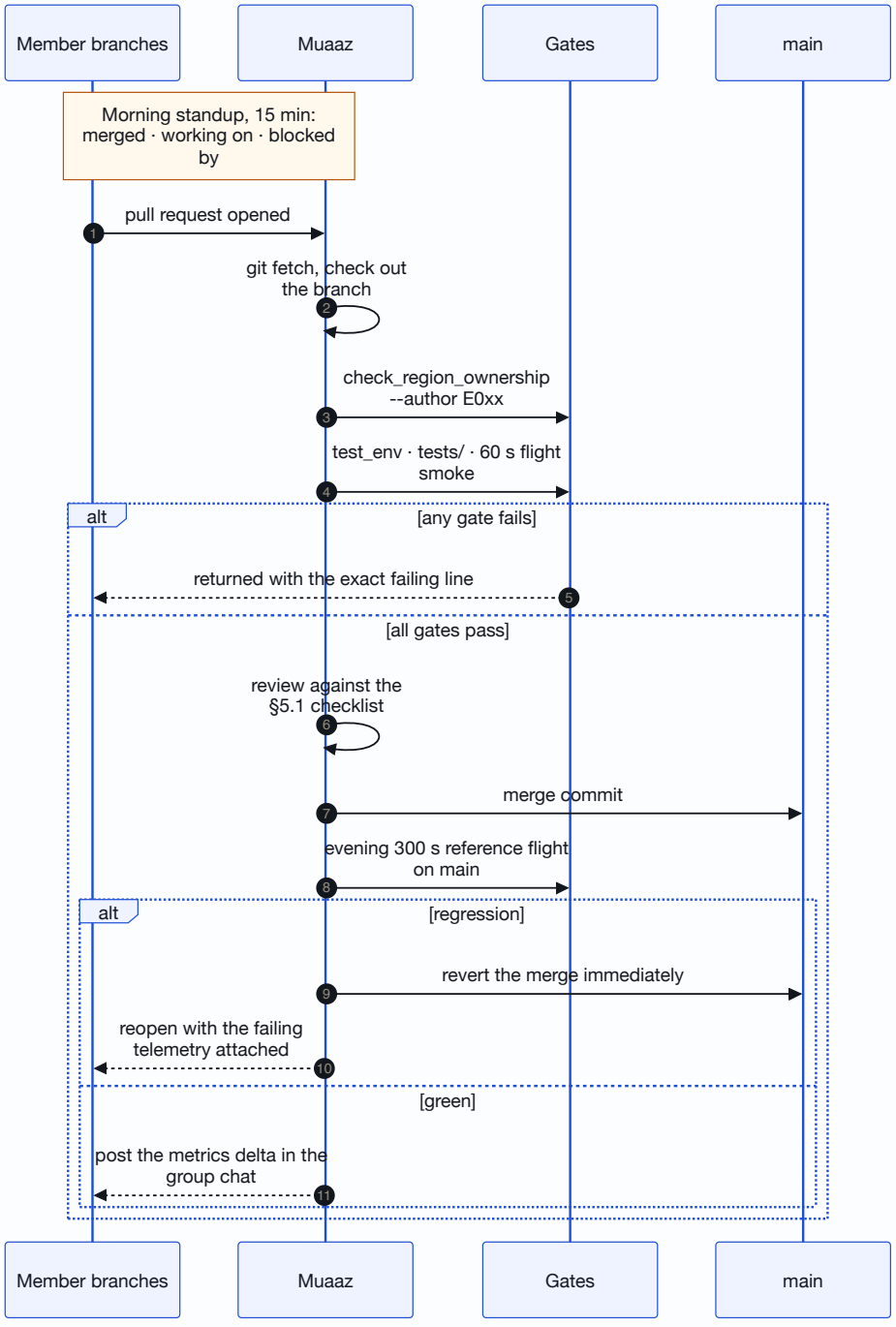


Figure 2 — Daily loop. Revert first, debug second: a broken `main` blocks three people.

5.1 Review checklist (cheapest rejections first)

#	CHECK	REJECT IF…
1	Region gate	`check_region_ownership.py` reports any line outside the author's region.
2	Frozen files	The diff touches `RESEARCH_AND_IMPLEMENTATION_GUIDE.md` or the generated `nmims_main_campus.xml` by hand.

#	CHECK	REJECT IF…
3	Smoke	`test_env.py`, `tests/`, or a 60 s flight fails.
4	Synthetic data	Any `np.random` appears outside the seeded gust field and intruder schedule, or a metric has a constant default.
5	Ground truth leakage	The detector reads `self.data`, `qpos` or any simulator state.
6	Hard-coded geometry	Zone bounds, ring points or building positions are typed in rather than read from the model.
7	Units and sources	A constant has no name, unit or source comment.
8	Honest stand-ins	Unfinished code returns a plausible value instead of raising or writing NaN.
9	Test	Behaviour changed and no test would have caught the old bug.
10	Authorship	The author cannot explain the change when asked.

6 Working in Parallel

Nobody waits for anybody, because every interface already exists in a form you can code against:

- **Aditya** — the loop already renders `cam_alpha` / `cam_bravo` at 640×480 and calls `detect_intruder_hsv`. Until M-09 lands, render frames yourself with `mujoco.Renderer` from the same model.
- **Zaid** — `self.substation_bounds` is already read from the model. The FSM and polygon test can be developed against plain coordinates in `tests/test_e075_*.py` with no detector at all.
- **Soumya** — the CSV schema is fixed and `analytics/runs/bench_smoke.csv` already has real rows. Write the economics against those columns now; it picks up new data automatically.

> **THE DAILY TEST**
> Ask each member “what are you waiting for?” If the answer names a person, the missing stub or fixture for that dependency is the highest-priority task on the team.

7 Backlog — E043 Muaaz FLIGHT · FLEET · INTEGRATION

ID	FEATURE	FR	EST.	DEFINITION OF DONE
M-01	NMIMS scene generator DONE	01	—	OSM footprints → MJCF; manifest; real-vs-sim figure.
M-02	Scaffold model fixes DONE	02–03	—	Nadir cameras, yaw torque, l = 0.26 m, f = 450 px; names preserved.
M-03	Physics loop, cascade, "+" mixer DONE	03–04	—	±0.033 m altitude; principal-frame inertia handled.
M-04	Layered separation DONE	05	—	Head-on regression test 3.29 m.
M-05	Sectors + offset ring DONE	06	—	58.5 s vs 41.0 s worst revisit.
M-06	Benchmark runner + provenance DONE	07	—	Course schema; NaN for pending columns.
M-07	Figure integrity + ownership gate DONE	22	—	Placeholder refused; gate self-tested.
M-08	**Initialise the git repo and branches**	—	0.5 d	Baseline commit on `main`; four course branches created; members cloned and committed once each.
M-09	Frame fixtures for Aditya	11	1 d	300 nadir frames at known intruder positions, both cameras, with camera pose and projected ground-truth boxes in `tests/fixtures/`.
M-10	Illumination tiers	10	1 d	Four lighting presets (1000/250/50/5 lx equivalents) selectable per trial; rendered samples committed.
M-11	RTL execution	09	1 d	On Zaid's trigger the aircraft climbs to transit altitude and lands on its dock; logged phase change.
M-12	Battery state-of-charge model	09	0.5 d	SoC from integrated rotor energy against a stated capacity, so the 20 % trigger has an input.
M-13	80-trial paired batch	07	1 d	Same seeds under both policies; provenance committed; runs unattended.
M-14	Paired statistics + figures	—	1 d	Paired t-tests with CI and effect size; Figures 1–3 regenerated with no watermark.
M-15	LinkedIn demo footage	—	0.5 d	60–90 s render from `control_room`, `chase_*` and `tour_viewpoint`; URL into `TEAM_ROSTER.json`.
M-16	**Integration duty**	all	½ d/day	Gates, reviews, merges, evening reference flight, metrics delta.

8 Backlog — E051 Aditya COMPUTER VISION

ID	FEATURE	FR	EST.	DEFINITION OF DONE
A-01	HSV detector	11	2 d	`detect_intruder_hsv`: HSV threshold, morphological open/close, contours, largest blob → (u, v, bbox). Uses the frame only.
A-02	**Pose-aware projection**	12	1.5 d	Intersect the pixel ray with the ground using the full camera pose (`data.cam_xpos`, `data.cam_xmat`). Report the error of the guide's level-camera formula against it — measured tilt makes that error 11–13 m at p95.

ID	FEATURE	FR	EST.	DEFINITION OF DONE
A-03	IoU against ground truth	13	1 d	Project the intruder cylinder into the image for a GT box; IoU per frame; median and IQR.
A-04	Illumination benchmark	14	1 d	`analytics/vision_illumination_benchmark.csv` : latency and IoU per lux tier, wall-clock timed.
A-05	False-alarm mitigation	—	1 d	The scene contains red tempo cabs and red signal lamps — a built-in false-positive test. Report the false-alarm rate before and after shape/size filtering.
A-06	Target hand-off between cameras	—	1 d	When the intruder leaves Alpha's footprint and enters Bravo's, one track continues.

BOUNDARY ADITYA MUST HOLD

The detector must never read simulator state. If it does, detection latency becomes a property of the simulator instead of the vision system, and objective O2 can no longer be falsified.

9 Backlog — E075 Zaid GEOFENCING & TELEMETRY

ID	FEATURE	FR	EST.	DEFINITION OF DONE
Z-01	**Use the new scene's zone**	15	0.5 d	Replace the scaffold's `(13, 15, 23, 25)` with `self.substation_bounds`, which is read from the model. Until then every tier is computed against a zone that no longer exists.
Z-02	Ray-casting point-in-polygon	15	1 d	General polygons, not just rectangles; tests for inside, outside, on-edge, on-vertex and concave cases.
Z-03	Hysteresis	16	1 d	Enter 0.5 m inside; clear only 2.0 m outside for 3 consecutive frames; a walk along the border produces no chatter.
Z-04	4-tier FSM incl. 5 m buffer	16	1 d	Tier transitions logged with time and aircraft.
Z-05	RTL triggers	17	0.5 d	SoC ≤ 20 % or heartbeat loss > 1.5 s raises the trigger Muaaz's M-11 executes.
Z-06	Alert packets	18	1 d	JSON/CSV with tier, time, aircraft and ground point.
Z-07	GPS/IMU noise injection	—	1 d	Configurable noise on the position used by the geofence; tier stability reported versus noise level.
Z-08	`blind_spot_pct`	19	1.5 d	Sample the wall line; mark a point covered when it falls inside an unoccluded camera footprint (ray-cast against trees and buildings); fill the CSV column.

10 Backlog — E077 Soumya CSBS OPERATIONS

ID	FEATURE	FR	EST.	DEFINITION OF DONE
S-01	**Define the guard comparator**	20	1 d	Written definition — perimeter-only walk, or full security round including floors — agreed with faculty. It decides whether the speedup is roughly 7× or 70×.
S-02	Guard baseline	20	1.5 d	`analytics/guard_patrol_baseline.py` fills `guard_cycle_min` and `guard_detect_lat_s` from a measured round or a stated walking model on the same 342 m ring and intrusion schedule.
S-03	Site facts	—	0.5 d	Storey count, substation location and guard roster from the NMIMS estates/security office; feeds the scene manifest.
S-04	Economics from the CSV	21	1 d	`campus_security_economics.py` reads cycle times and latencies from the benchmark CSV; no typed-in results remain.
S-05	Sourced normalised costs	21	1 d	C_{hybrid} and K_{CapEx} argued from stated assumptions, or swept parametrically.
S-06	Payback band	21	1 d	Payback across the assumption range, with the break-even point stated.
S-07	Manuscript tables	—	1 d	Every table generated from the CSV; none typed by hand.

11 Dependencies & Merge Order

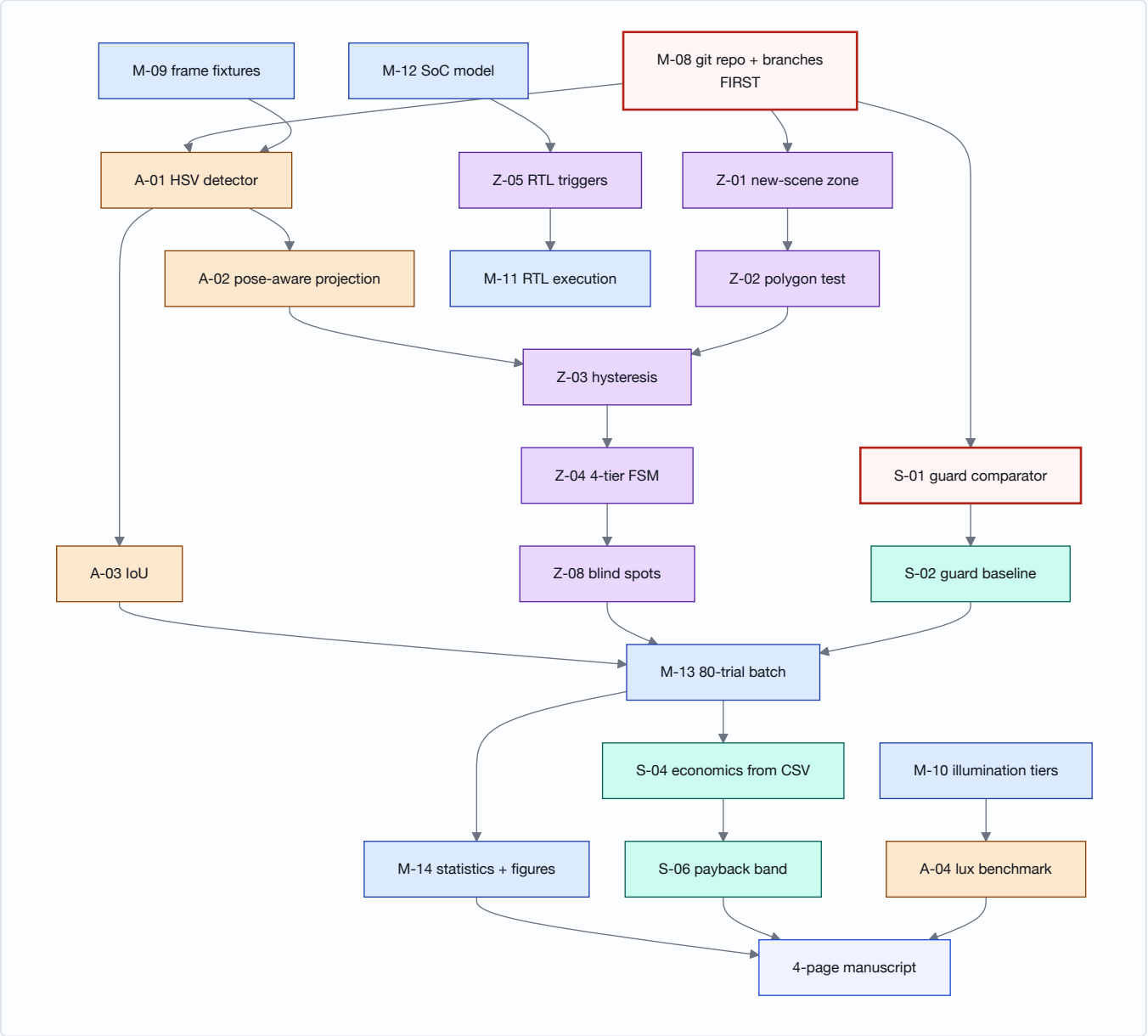


Figure 3 — Dependencies. Red items gate everything: without the repository no one can open a pull request, and without the guard comparator the headline speedup has no denominator.

1. **Fixes before features.** Z-01 merges before any FSM work — it removes the stale zone.
2. **Projection before tiers.** A-02 merges before Z-03, or the hysteresis is tuned against metres of projection error.
3. **Scene changes are serialised.** Only Muaaz edits `models/`; others request changes.
4. **Batch last.** M-13 runs only after A-03, Z-08 and S-02 are merged, so no column is NaN.

12 Sprint Schedule

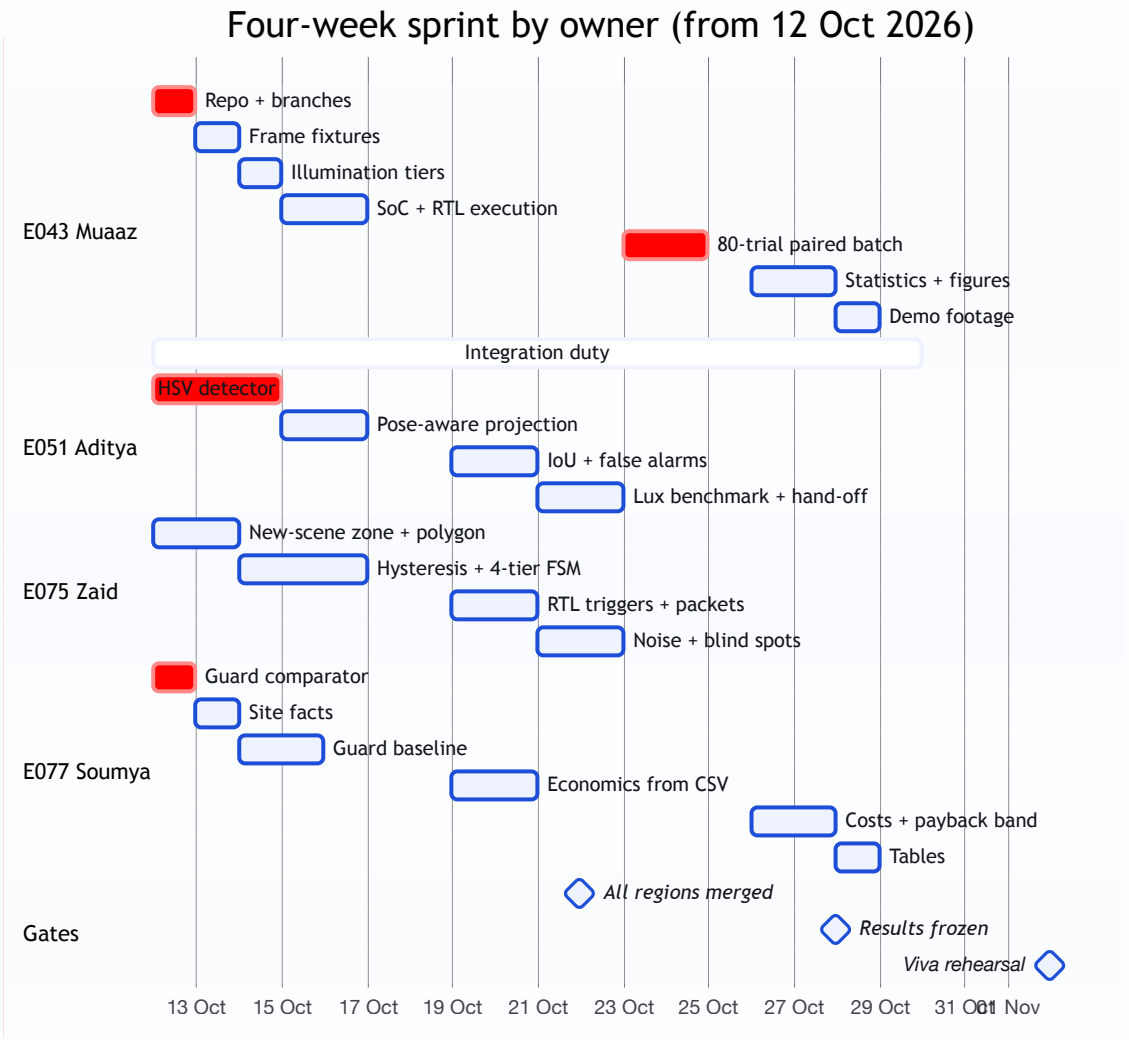


Figure 4 — Schedule. Every member has productive work on day one. Pin the dates to the academic calendar.

13 Quality Gates

```
# before every merge, run by Muaaz on the PR branch
python src/check_region_ownership.py --author E0xx      # line-level ownership
python src/test_env.py                                  # course toolchain check
python tests/test_e043_separation.py                    # head-on separation >= 2.5 m
python -m pytest tests -q                               # everyone's tests
python src/aerial_patrol_swarm.py --duration 60 --tag ci --no-vision
python analytics/generate_paper_figures.py              # must refuse placeholder data (exit 2)
```

GATE	WHEN	PASSES IF
Pre-merge	Every PR	All commands above succeed (the last one by refusing).
Evening reference flight	Daily on `main`	300 s, zero collisions, altitude error under 0.1 m, separation ≥ 2.5 m.
Regions merged	22 Oct	No `NotImplementedError` remains; no CSV column is NaN in a smoke batch.
Evidence audit	28 Oct	Fresh clone → `run_benchmark.py` → figures and tables regenerate identically.

14 Definition of Done

1. The FR's verification passes and its status in PRD §8 is updated in the same PR.
2. Every changed line lies in the author's region or owned files (gate passes).
3. A test fails if the feature regresses.
4. Any number it produces lands in a CSV via `run_benchmark.py`.
5. Constants are named, carry units and cite a source.
6. The author can explain every line, unprepared, in two minutes.
7. Merged to `main`, and the evening reference flight is green afterwards.

15 Contingencies

IF…	DECIDE BY	THEN
The detector isn't working by 19 Oct	19 Oct	Publish patrol-cycle and separation results (O1, O3, O4) — they stand alone — and report detection as in-progress. Never substitute geometric visibility for detection.
Site facts (storeys, substation) unobtainable	16 Oct	Keep the assumed values, state them in the methods section, and run a sensitivity check on building height.
Faculty rejects the NMIMS scene	On reply	The generator can emit the scaffold's 60 × 60 m campus; the flight stack is scene-agnostic because it reads checkpoints and zones from the model.
Batch does not fit the calendar	21 Oct	Cut illumination tiers to two before cutting seeds — seeds are what produce confidence intervals.
A member becomes unavailable	Immediately	Cut their scope to the Must items and say so in the paper. Their region is not completed under another name.
Offset ring does not beat sectors significantly	27 Oct	Report it. A well-instrumented null result is publishable; a massaged one is a course failure.

16 Viva Preparation

The guide supplies model answers. Two of Muaaz's are contradicted by measurement, and saying so with evidence is a stronger answer than repeating them.

MEMBER	LIKELY QUESTION	ANSWER GROUNDED IN THIS REPOSITORY
E043	Does your APF guarantee d ≥ 2.5 m at the sector hand-off?	Not on its own. Measured head-on at 6 m/s each: 0.40 m — contact. The closing energy is 48.6 J; the field can only absorb it by d = 0.26 m, and rotor thrust saturates long before the force "diverges". Separation therefore comes from altitude layers, a closing-speed governor and a right-hand sidestep, with APF as the inner layer — 3.29 m in the same test.
E043	How does yaw control work with four upward thrusts?	Through rotor drag torque: opposite pairs spin opposite ways, encoded as $\pm k_Q$ in each rotor's gear vector. The scaffold had $k_Q = 0$, which gives no yaw authority at all.
E043	Why read inertia carefully?	MuJoCo reports it in the principal frame, reordered here so I_{zz} comes first; using it raw would control roll with the yaw inertia.
E051	Derive the inverse pinhole projection — when is it valid?	Only for a level, North-aligned camera. In flight the tilt is 7° median and 28° at p95, giving 11–13 m error at p95; the ray must be intersected with the ground using the full camera pose.
E051	Would HSV work on a real person?	No — the avatar is red by design. The scene's red tempo cabs and signal lamps measure how quickly colour-only detection produces false alarms.
E075	How does hysteresis prevent chatter?	Enter at 0.5 m inside, clear only 2.0 m outside for 3 frames; demonstrated by a test walk along the border.
E075	How is blind-spot percentage computed?	Wall-line samples ray-cast from each camera with tree and building occlusion — not a typed-in figure.
E077	Is a 6.21× speedup real?	It was a placeholder. The measured value depends on the comparator definition, which is stated in the paper.
E077	How is payback computed without currency?	Normalised to the guard-only budget ≡ 1.00, inputs from the CSV, reported as a band with the break-even point.

Rehearsal (2 Nov): each member is questioned for ten minutes by the other three, live-editing their own region and re-running it. Muaaz goes last and is questioned on the whole system.

Implementation & Review Plan v2.0 · companion: *PRD v2.0*.